Sharpness, Robustness Penalties, and Hessian Geometry

Fiber Raman Suppression Project

Evidence snapshot: 2026-04-26

Abstract

This note asks whether the default Raman-suppression optimizer should be replaced by an explicitly robustness-aware objective. The tested alternatives were Monte Carlo averaging over phase perturbations, sharpness-aware minimization, and a finite-difference Hessian-trace penalty. The main result is negative but useful: the plain log-cost optimizer remains the best default when the goal is maximum Raman suppression, while Monte Carlo and trace penalties are optional tools when tolerance to phase-shaper error matters more than depth. The geometry result is also important: every resolved Hessian spectrum in this sweep stayed indefinite, so these solutions are better understood as saddle-like stationary points rather than clean convex minima.

Claim Status

Status: established tradeoff; not a default replacement
The sweep supports a stable conclusion: robustness penalties can buy perturbation tolerance, but they cost Raman depth and do not remove saddle-like curvature. The plain objective remains the default. The robustness objectives are useful research knobs, not a new universal optimizer.

1 Question and Thesis

The baseline note defines the usual objective: choose a spectral phase $\phi(\omega)$, propagate the pulse through the fiber, and minimize the Raman-band output fraction. This note asks a different question:

Can we optimize not only for a deep result, but for a less fragile result?

The motivation is practical. A phase mask that gives excellent suppression in simulation may be hard to use experimentally if small phase errors ruin it. A more robust mask might tolerate small shaper errors, calibration drift, or minor parameter mismatch. In this project, robustness is measured by how large a random phase perturbation can be before the solution loses 3 dB of suppression.

The thesis is:

robustness can be improved, but it is not free.

The strongest trace-penalty results widen the perturbation tolerance, but they also give much shallower Raman suppression. Monte Carlo averaging gives a smaller but cheaper tolerance improvement. Sharpness-aware minimization did not produce a useful Pareto improvement in this sweep. None of the variants turned the local Hessian into a positive-definite minimum.

Intuition Check

The plain optimizer asks, "how low can the Raman objective go?" The robust objectives ask, "can the result stay good if the phase is nudged?" These are not the same question. A very deep optimum can be narrow and fragile, while a shallower optimum can be wider and easier to tolerate.

2 Common Setup

All results in this note use the same phase-only control idea as the baseline:

$$u_\omega(0;\phi) = u_{\omega,0}\exp(i\phi(\omega)).$$

The output Raman objective is the Raman-band energy fraction

$$J_{\mathrm{phys}}(\phi) = \frac{\sum_{\omega_k\in\mathcal{B}_{\mathrm{Raman}}} |u_{\omega_k}(L;\phi)|^2\,\Delta\omega}{\sum_k |u_{\omega_k}(L;\phi)|^2\,\Delta\omega},$$

reported in decibels as

$$J_{\mathrm{dB}}(\phi) = 10\log_{10} J_{\mathrm{phys}}(\phi).$$

Lower J_{dB} is deeper suppression. The forward propagation follows the standard GNLSE modeling stack for nonlinear fiber optics [1, 2, 3]. L-BFGS is used for the first-order optimization step [4, 5].

Two operating points were used. The first is a full-grid SMF-28 point at $L = 0.5\,\mathrm{m}$, $P = 0.05\,\mathrm{W}$, and $N_t = 8192$. The second is a reduced-control SMF-28 point with 57 cubic-basis coefficients. The reduced-control point is included because a robustness method should be tested both on the direct full-grid parameterization and on a structured control space.

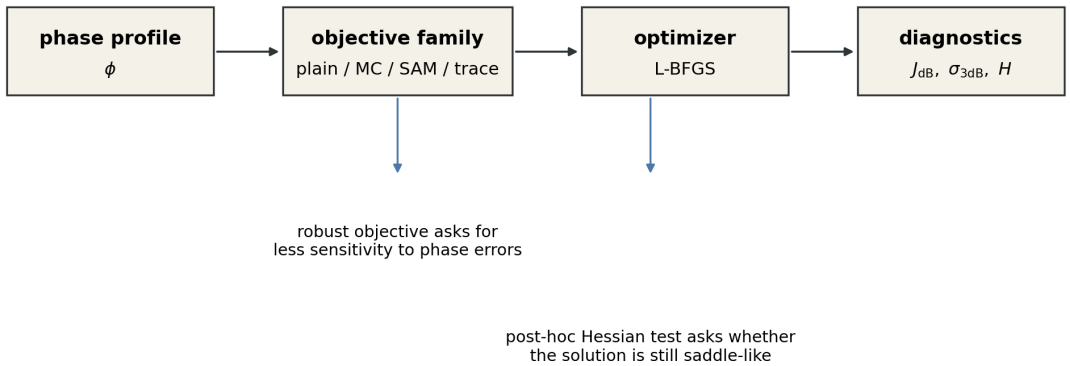

Figure 1: Sharpness/robustness workflow. The physical propagation model is the same as the baseline. This research lane changes the objective family and then diagnoses the resulting solution with suppression depth, perturbation tolerance, and Hessian curvature.

3 Robust Objective Variants

Let $F(\phi)$ denote the log-scale Raman objective optimized by L-BFGS. The plain baseline solves

$$\min_{\phi} F(\phi).$$

The robustness variants modify the scalar objective before L-BFGS sees it.

3.1 Monte Carlo Phase Averaging

The Monte Carlo objective minimizes expected performance under random phase error:

$$F_{\mathrm{MC}}(\phi) = \frac{1}{K}\sum_{k=1}^{K} F(\phi + \epsilon_k), \qquad \epsilon_k \sim \mathcal{N}(0, \sigma^2 I).$$

In the implementation, the perturbation bank is fixed during one optimization run and uses antithetic pairs $(+\epsilon, -\epsilon)$. This matters because L-BFGS expects a deterministic objective during line search. If new random noise were drawn every call, the optimizer would chase a moving target.

Intuition Check

Monte Carlo averaging is the most literal robustness objective: instead of grading one exact phase mask, we grade nearby masks too. If a solution only works at one razor-thin phase setting, this averaged objective punishes it.

3.2 Sharpness-Aware Perturbed Loss

Sharpness-aware minimization (SAM) approximately solves a local min-max problem:

$$F_{\mathrm{SAM}}(\phi) = F\left(\phi + \rho \frac{\nabla_\phi F(\phi)}{\|\nabla_\phi F(\phi)\|}\right).$$

The perturbation direction is the local steepest direction in phase space, and ρ sets the neighborhood radius. The idea comes from sharpness-aware training in machine learning [6]. Related work warns that sharpness measures can be scale-sensitive or misleading, so this note treats SAM as one robustness probe, not as the definition of flatness [7, 8].

Intuition Check

SAM asks, "if I step in the direction that locally makes the cost worse, is the result still okay?" That is a useful stress test, but it is not identical to real shaper noise, and it does not guarantee that the Hessian is flat.

3.3 Hessian-Trace Penalty

The trace-penalty objective adds a finite-difference estimate of local curvature:

$$F_{\mathrm{trace}}(\phi) = F(\phi) + \lambda_{\mathrm{tr}} S(\phi),$$

where

$$S(\phi) \approx \frac{1}{N_s} \sum_{i=1}^{N_s} \frac{F(\phi + \epsilon P v_i) + F(\phi - \epsilon P v_i) - 2F(\phi)}{\epsilon^2}.$$

Here v_i are fixed Rademacher probe vectors, and P projects out two gauge directions: constant phase and linear phase over the input band. Those directions do not represent meaningful Raman-control changes, so the sharpness estimator should not spend probes measuring them. Hutchinson-style trace estimation is a standard way to estimate matrix traces without forming the matrix explicitly [9, 10].

Intuition Check

The trace penalty is trying to punish directions where the objective bends sharply. But the Raman landscape is not a simple bowl. It has both upward and downward curvature directions, so a signed trace can hide cancellation. That is why the note also checks Hessian eigenvalues after optimization.

4 Robustness and Hessian Diagnostics

The main robustness metric is $\sigma_{3\mathrm{dB}}$. Starting from an optimized phase $\phi_\star$, draw random phase perturbations of size σ and evaluate the resulting suppression. The value $\sigma_{3\mathrm{dB}}$ is the approximate perturbation amplitude where the solution loses 3 dB relative to its unperturbed suppression:

$$J_{\mathrm{dB}}(\phi_\star + \delta_\sigma) \approx J_{\mathrm{dB}}(\phi_\star) + 3\,\mathrm{dB}.$$

Larger $\sigma_{3\mathrm{dB}}$ means the phase mask is more tolerant to random phase error.

After optimization, the sweep also measured Hessian spectra. Let

$$H(\phi_\star) = \nabla_\phi^2 F(\phi_\star)$$

or the corresponding control-space Hessian for reduced-control runs. A positive-definite Hessian would mean every local direction curves upward. An indefinite Hessian means there are both upward and downward curvature directions: locally, the point is saddle-like.

Hessian-vector products were computed by finite differences of the gradient:

$$Hv \approx \frac{\nabla F(\phi_\star + \epsilon v) - \nabla F(\phi_\star - \epsilon v)}{2\epsilon}.$$

This avoids forming the full Hessian matrix. The eigenspectrum check is still post-hoc: it diagnoses geometry after the optimization, rather than changing the L-BFGS step.

Intuition Check

Think of $\sigma_{3\mathrm{dB}}$ as the practical tolerance test and the Hessian as the geometry test. The tolerance test asks, "does the mask survive random phase noise?" The Hessian test asks, "is the nearby landscape shaped like a bowl or like a saddle?" This sweep says the masks can become more tolerant, but the geometry still looks saddle-like.

5 Implementation Surface

Item	Role in this note
`scripts/lib/sharpness_optimization.jl`	Shared sharpness helper for gauge projection and Hutchinson-style finite difference trace penalties.
`scripts/research/sharpness/lib.jl`	Research library for objective variants, operating-point setup, perturbation banks, robustness scans, and Hessian diagnostics.
`scripts/research/sharpness/run.jl`	Batch driver for the sharpness/robustness sweep.
`scripts/research/sharpness/summarize.jl`	Post-processing script that produced the robustness-depth table and summary used here.
`scripts/lib/standard_images.jl`	Standard phase diagnostic and evolution heat-map generator used for the representative profiles.

Table 1: Main implementation surface for the sharpness/robustness note.

6 Experimental Strategy

The sweep compared four objective families:

plain, Monte Carlo, SAM, trace penalty.

For each successful solution, the post-processing step recorded:

- Raman depth J_{dB}, where more negative is better.
- Perturbation tolerance $\sigma_{3\mathrm{dB}}$, where larger is better.
- Hessian status, including whether the resolved spectrum was indefinite.
- Standard phase and evolution images for representative cases.

The sweep completed 26 successful records and 0 failed records. All 26 resolved Hessian spectra were indefinite. That all-indefinite result is the main geometry conclusion.

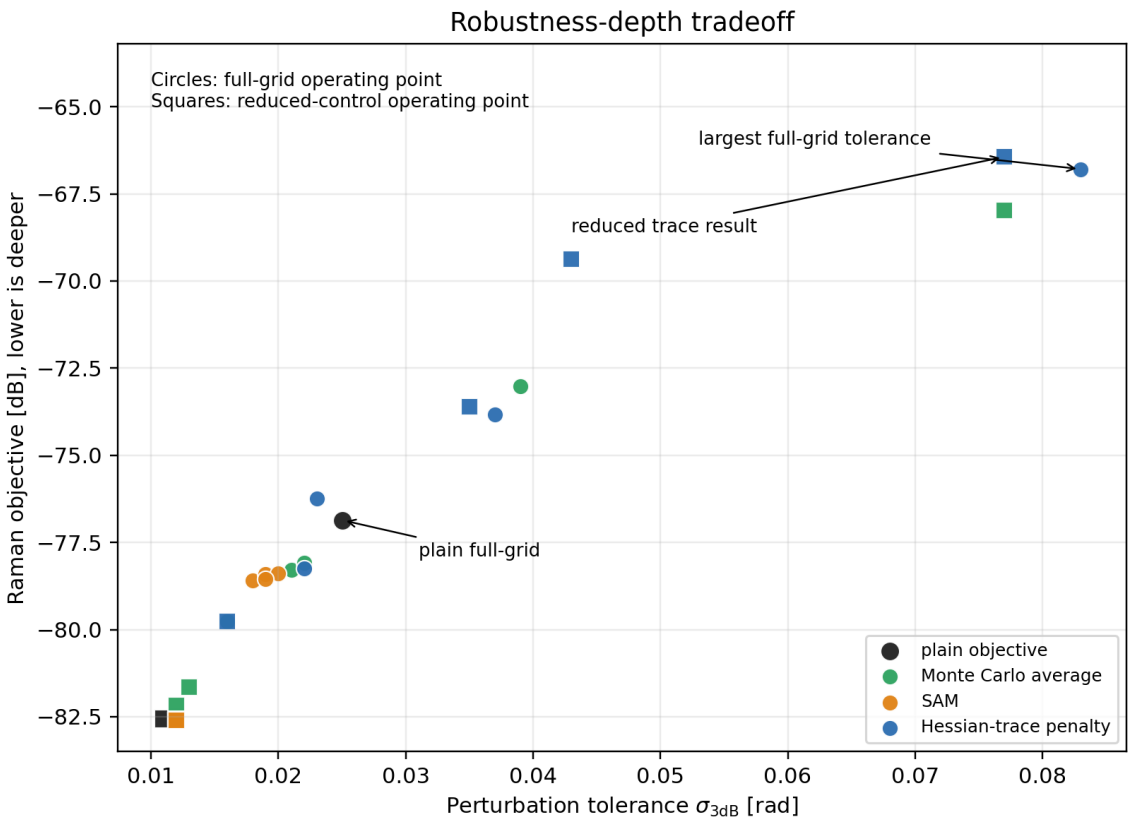

Figure 2: Depth versus perturbation tolerance. Moving right means more tolerance to phase noise. Moving down means deeper Raman suppression. The strongest trace-penalty cases move far to the right but lose suppression depth, while the Monte Carlo cases give smaller tolerance gains.

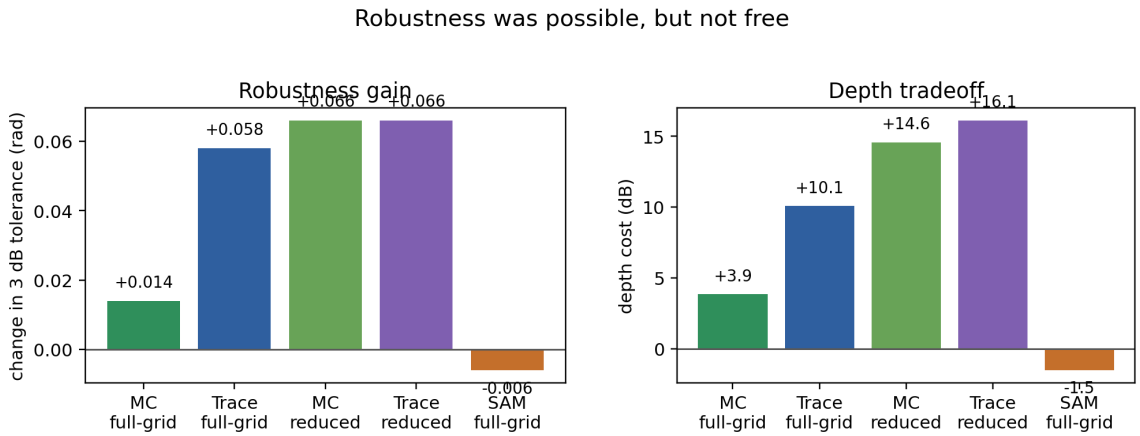

Figure 3: Robustness gain versus depth cost for representative methods. Monte Carlo gives the cheaper full-grid improvement. The trace penalty gives the largest tolerance gain, but the depth loss is large enough that it should not replace the default objective.

Interpretive Summary

This is a tradeoff plot, not a leaderboard with one universal winner. A point near the lower left is deep but fragile. A point near the upper right is more tolerant but shallower. For the current project default, depth matters enough that the plain objective remains the reference.

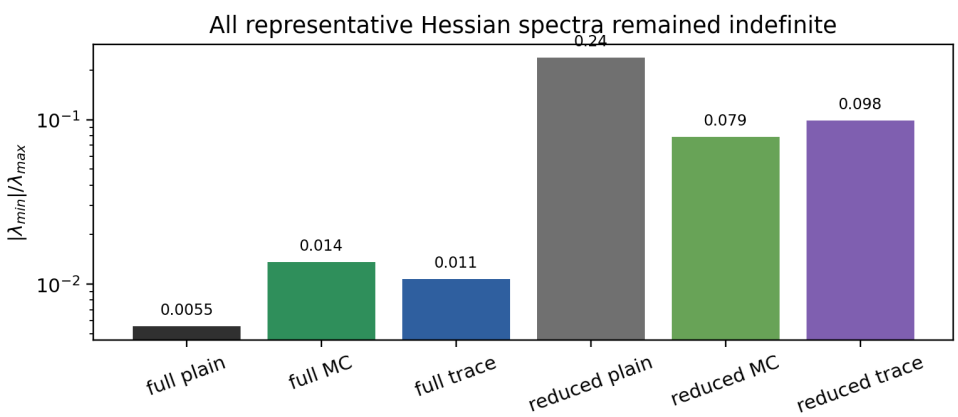

Figure 4: Representative Hessian indefiniteness ratios. Each shown spectrum has both positive and negative curvature. Robustness penalties changed tolerance, but they did not turn the tested optima into positive-definite local minima.

7 Representative Results

The following pages use the standard research-note format: first the phase diagnostic, then the matching spectral-evolution heat map on the same page. Read each robustness case against the no-optimization control and the plain objective reference.

How To Read These Pages

Phase diagnostic. The diagnostic shows the applied phase, unwrapped phase, group delay, group-delay dispersion, and instantaneous frequency shift. Large spikes or rapid oscillations are warning signs that a mask may be difficult to implement experimentally.

Spectral-evolution heat map. The heat map shows how spectral power changes along the fiber. It is the best quick visual check for whether a phase mask changed the physical evolution or merely produced a different-looking input phase.

Control comparison. The no-optimization page is the control group. The plain objective page is the depth reference. The Monte Carlo and trace pages should be judged by the depth-versus-tolerance tradeoff, not by depth alone.

Intuition Check

The phase diagnostic is the input story. The heat map is the fiber-response story. A robustness claim needs both, plus the scalar tolerance number.

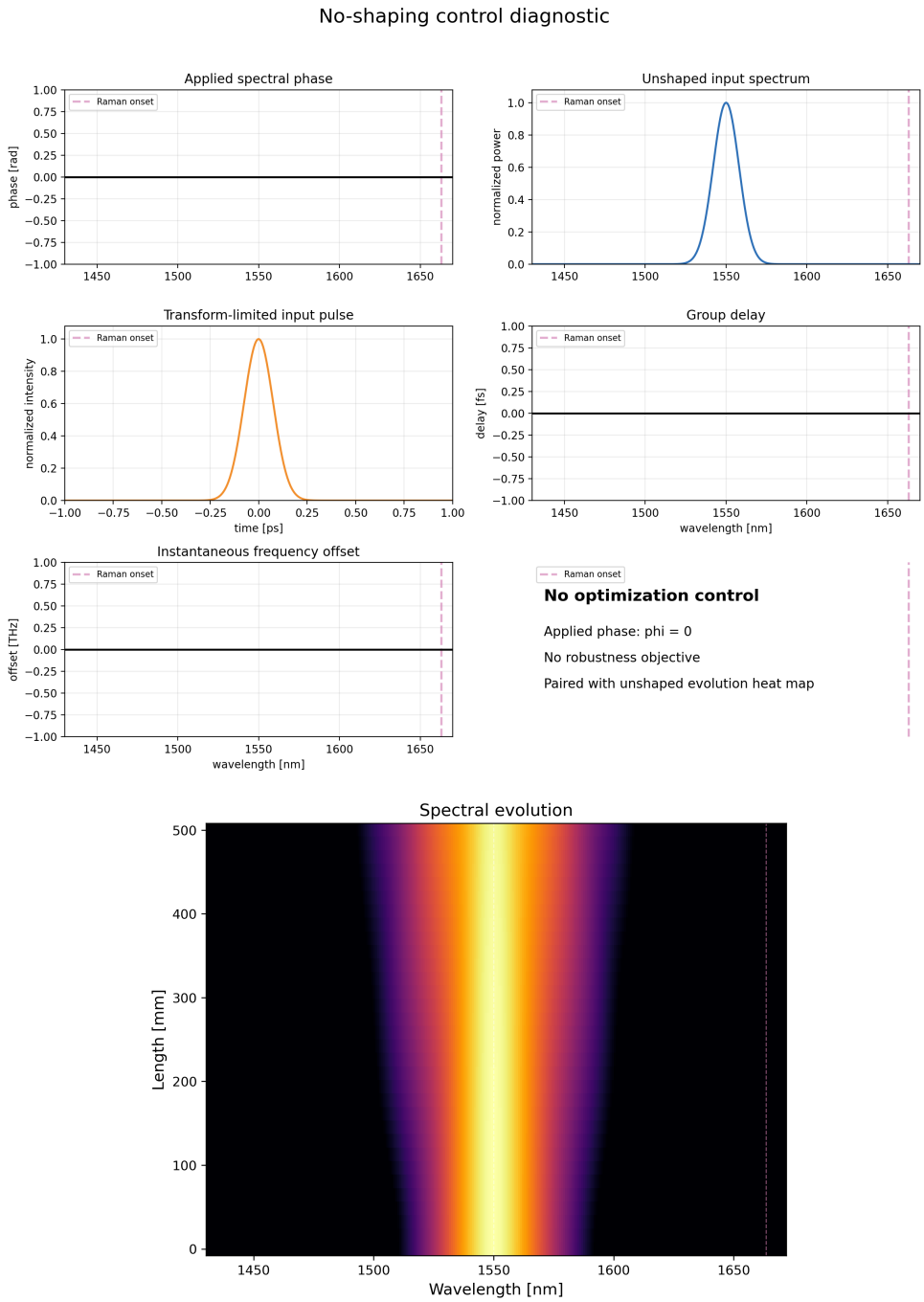

Figure 5: No-optimization control. The left diagnostic has flat applied phase; the heat map shows the unshaped propagation for the full-grid operating point. This page is the reference before any sharpness or robustness objective is allowed to change the phase.

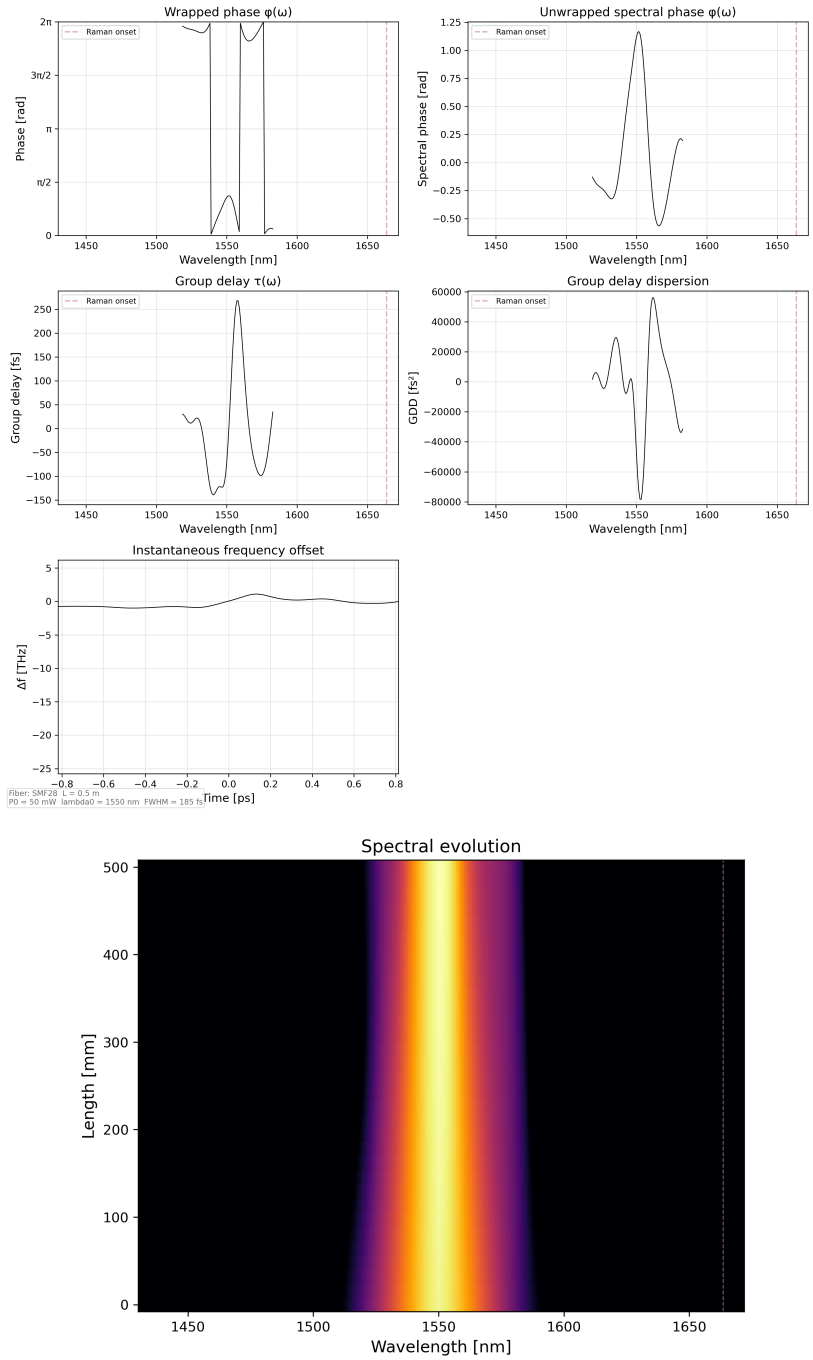

Figure 6: Plain full-grid objective. This solution reached about $-76.86\,\mathrm{dB}$ with $\sigma_{3\mathrm{dB}} \approx 0.025\,\mathrm{rad}$. It is the depth-oriented reference case for this note.

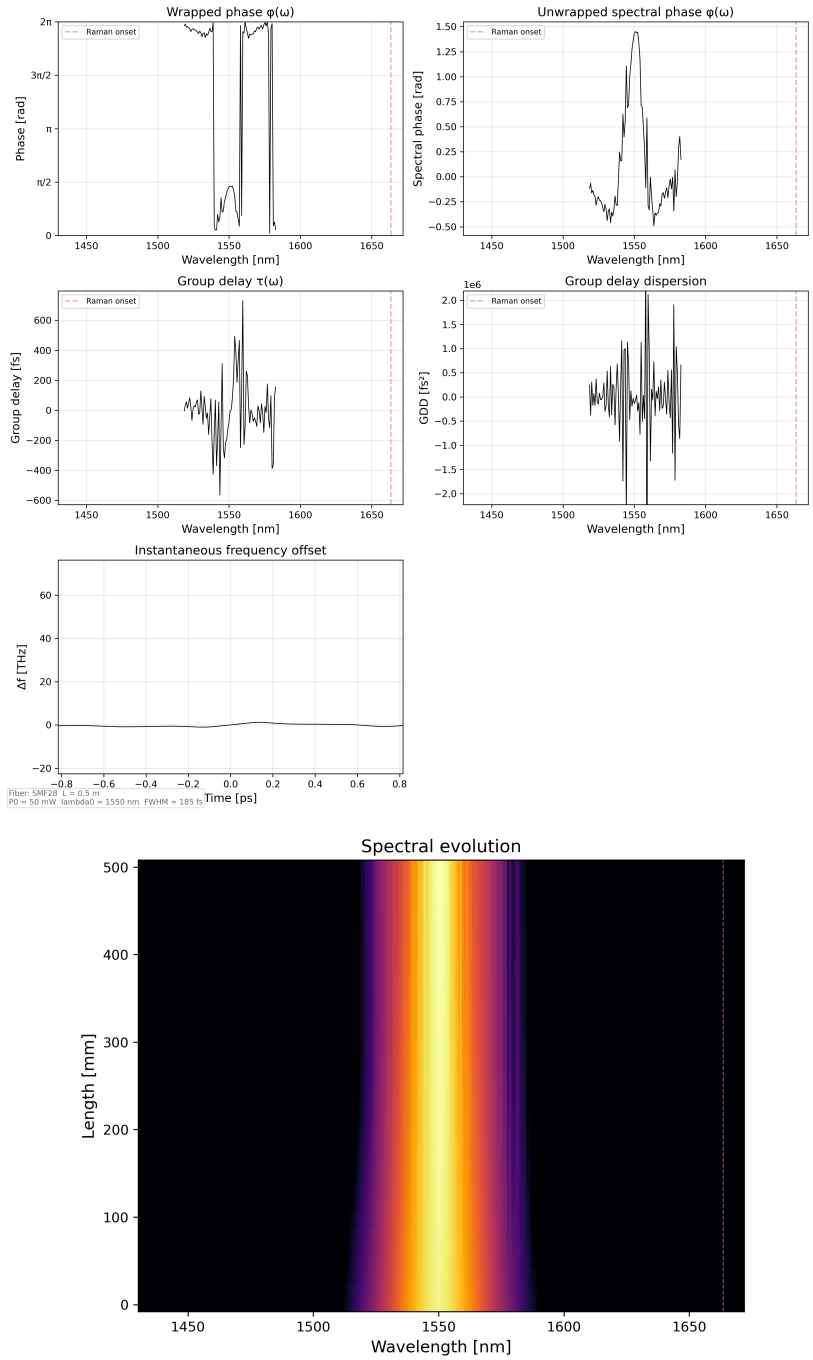

Figure 7: Monte Carlo phase-averaged objective on the full-grid operating point. The strongest tested MC case reached about $-73.01\,\mathrm{dB}$ with $\sigma_{3\mathrm{dB}} \approx 0.039\,\mathrm{rad}$, improving tolerance by about $0.014\,\mathrm{rad}$ at a depth cost of about $3.85\,\mathrm{dB}$.

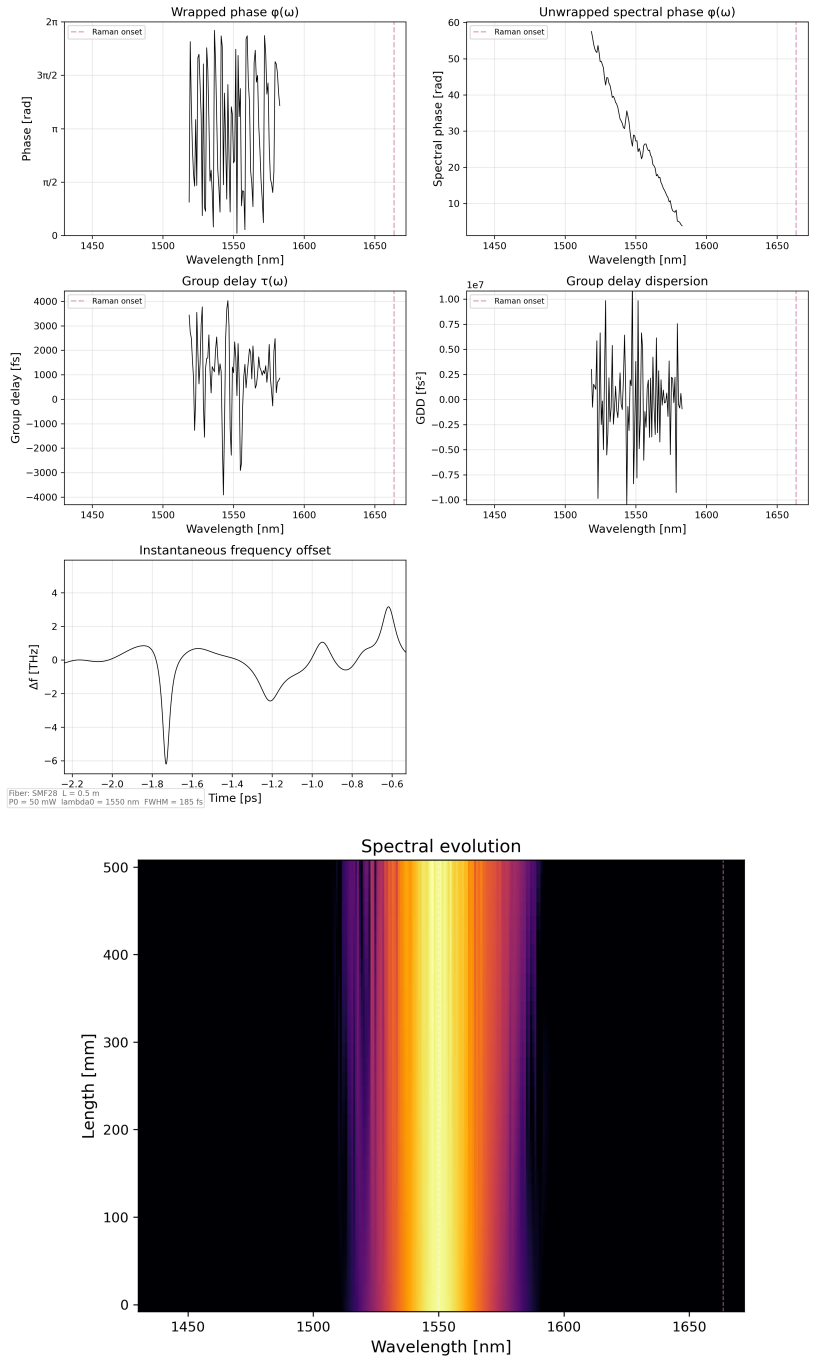

Figure 8: Hessian-trace penalty on the full-grid operating point. This case reached about $-66.79\,\mathrm{dB}$ with $\sigma_{3\mathrm{dB}} \approx 0.083\,\mathrm{rad}$. It gives the largest full-grid tolerance gain, but the depth cost is about $10.08\,\mathrm{dB}$.

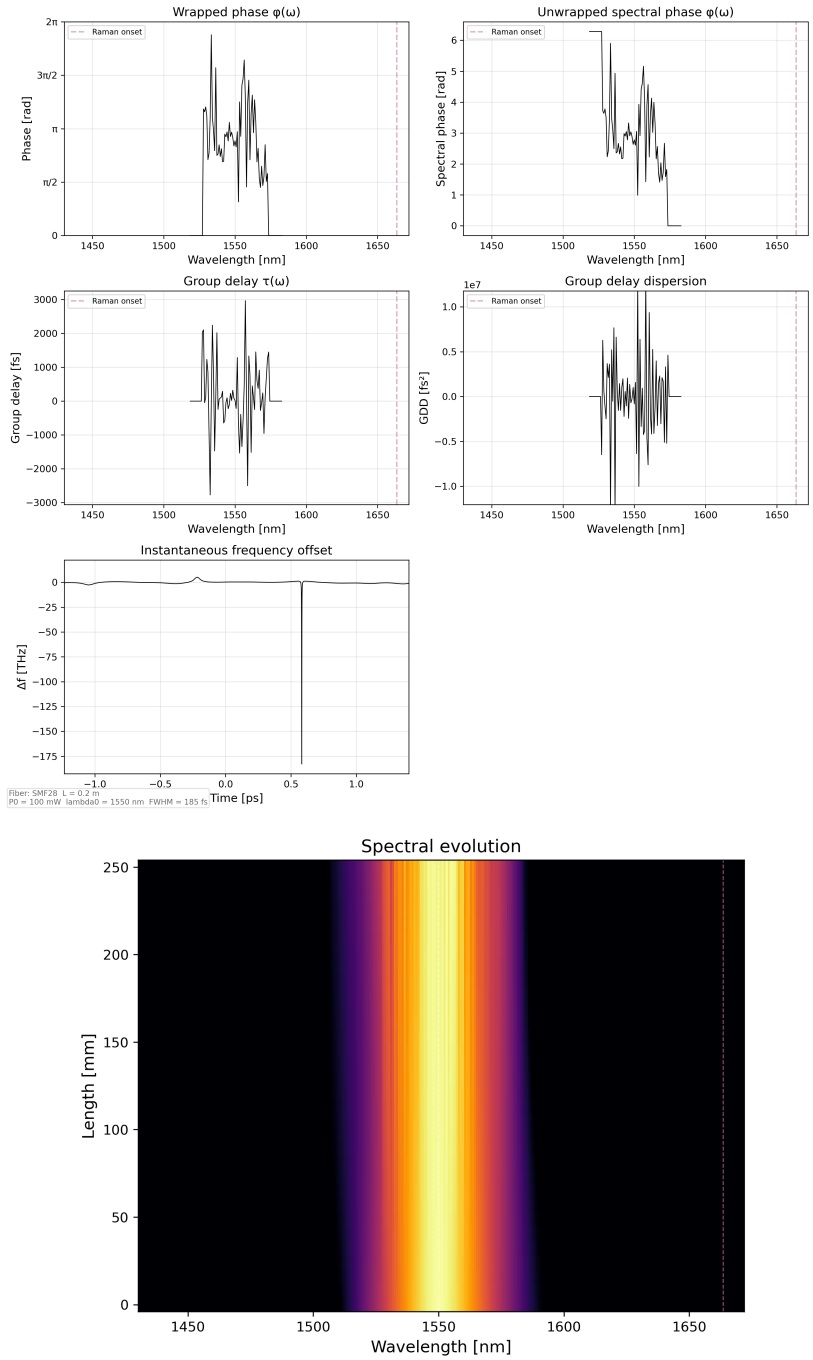

Figure 9: Trace penalty on the reduced-control operating point. The best reduced-control tolerance result reached about $-66.44\,\mathrm{dB}$ with $\sigma_{3\mathrm{dB}} \approx 0.077\,\mathrm{rad}$. This confirms that the robustness-depth tradeoff is not only a full-grid artifact.

8 Key Numerical Results

Operating point	Objective	Raman depth	$\sigma_{3\mathrm{dB}}$	Geometry
Full-grid	plain	$-76.86\,\mathrm{dB}$	$0.025\,\mathrm{rad}$	indefinite
Full-grid	MC, strongest tolerance	$-73.01\,\mathrm{dB}$	$0.039\,\mathrm{rad}$	indefinite
Full-grid	SAM, best tested	$-78.38\,\mathrm{dB}$	$0.020\,\mathrm{rad}$	indefinite
Full-grid	trace, strongest tolerance	$-66.79\,\mathrm{dB}$	$0.083\,\mathrm{rad}$	indefinite
Reduced-control	plain	$-82.56\,\mathrm{dB}$	$0.011\,\mathrm{rad}$	indefinite
Reduced-control	MC, strongest tolerance	$-67.98\,\mathrm{dB}$	$0.077\,\mathrm{rad}$	indefinite
Reduced-control	trace, strongest tolerance	$-66.44\,\mathrm{dB}$	$0.077\,\mathrm{rad}$	indefinite

Table 2: Representative numerical results from the sharpness/robustness sweep. The table keeps only the comparison points needed to interpret the tradeoff; the full sweep contained 26 successful records.

9 Interpretation

The first lesson is that robustness penalties do work in the limited sense that they can increase $\sigma_{3\mathrm{dB}}$. The strongest full-grid trace case moved from $0.025\,\mathrm{rad}$ to $0.083\,\mathrm{rad}$, and the strongest reduced-control cases reached about $0.077\,\mathrm{rad}$.

The second lesson is that the price is large. The strongest full-grid trace case lost about $10\,\mathrm{dB}$ of Raman depth. The reduced-control high-tolerance cases lost even more depth relative to their very deep plain reference. That is why the default objective should not be replaced.

The third lesson is that "robust" and "convex-looking" are different claims. All resolved Hessian spectra remained indefinite. In other words, the robustness penalties widened perturbation tolerance in some cases, but they did not turn the local landscape into a simple bowl.

TLDR

If I had to explain the result quickly: the sharpness lane found useful knobs, not a new default. Trace regularization can make the phase less fragile, MC can help a little more cheaply, SAM did not help much here, and the Hessian still says the landscape is saddle-like.

10 Presentation Capsule

If this note becomes a robustness slide, the clean message is:

- Deep suppression and robust suppression are different axes; the deepest mask can be experimentally fragile.
- Sharpness penalties intentionally trade some Raman depth for larger tolerance to phase perturbations.

- The most useful visual is the depth-versus-robustness tradeoff, not a short-fiber before/after image.
- The Hessian evidence still shows saddle-like geometry, so robustness penalties are knobs, not a proof that the landscape became simple.

11 Limitations

The robustness metric uses random phase perturbations, which are a useful proxy for shaper error but not a complete model of an experimental pulse shaper. Real hardware errors can have wavelength-dependent structure, calibration offsets, amplitude coupling, finite resolution, and repeatability limits.

The Hessian diagnostics are finite-difference diagnostics around the final solution. They are not a second-order optimizer, and the result depends on the chosen scalar objective and control space. In particular, the reduced-control Hessian should be read as curvature in coefficient space, not as a claim about every full-grid direction.

The trace penalty uses a stochastic trace estimator with fixed probes. That is the right deterministic choice for L-BFGS, but it still estimates only a sampled curvature quantity. Increasing probe count or testing structured hardware-like perturbations would be useful follow-up work.

12 Reproduction Capsule

The current sharpness research drivers live under `scripts/research/sharpness/`. The expected workflow is:

1. Run the batch driver with Julia threading enabled:

```
julia --project=.  -t auto scripts/research/sharpness/run.jl
```

2. Summarize the completed sweep:

```
julia --project=.  -t auto scripts/research/sharpness/summarize.jl
```

3. Inspect representative standard images before interpreting the results.

Substantial reruns belong on the burst machine using the project heavy-job wrapper. The important outputs are the completed record bundle, the robustness-depth plot, the summary table, and the standard image sets for representative plain, MC, SAM, and trace cases.

13 Sources and References

References

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